Evidence-Aware Reduction for Forkable Sandboxes

Separating Execution Fan-Out from Independent Evidence

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Abstract

Snapshot-backed sandboxes make branching cheap; they do not create independent evidence. Branches can reuse a model, prompt, repository, tests, observations, or execution ancestor, so counting outputs can turn one repeated error into high-confidence consensus. We introduce an *evidence-aware reduction contract*: each worker reports an estimate, estimated information, evidence identifiers, fork lineage, and execution metadata. For workers estimating a common parameter from disjoint independent evidence, the statistical core is the Gaussian/Wald common-effect case of established inverse-information and confidence-distribution pooling. Its numeric canonical state merges associatively and commutatively, while provenance follows explicit field semantics. Completing the square yields the pooled kernel and a residual heterogeneity statistic Δ—Cochran's Q in the scalar inverse-variance case—that also enters the product integral. A reference implementation validates serialized records, rejects repeated nonempty evidence identifiers, carries evidence and lineage through tree reduction, and uses Cholesky-based numerical linear algebra. Unit tests and seeded synthetic checks exercise the algebra, unequal information, and forged precision; one four-worker named-snapshot trace exercises the end-to-end path. Platform logs document the exercised execution paths. The next systems problem is to turn evidence identity and fork lineage into a dependence model for correlated and adaptively selected AI branches. Artifact: github.com/zozo123/boltzmann-mapreduce.

1 Introduction

Execution fan-out and evidence fan-out are different resources. A snapshot runtime can restore one prepared environment into many isolated workers, but shared models, prompts, tests, data, repositories, and ancestors can induce common-mode error. If K scalar estimates have equal marginal variance σ^2 and exchangeable pairwise correlation ρ, then

$$\operatorname{Var}(\bar{\theta}) = \sigma^2 \left[\rho + \frac{1-\rho}{K}\right], \tag{1}$$

whereas an independence calculation reports σ^2/K. For fixed $\rho > 0$, variance has the floor $\rho\sigma^2$ even as the number of branches grows. Execution isolation constrains runtime state; it does not determine ρ.

MapReduce gained leverage from a small interface that let the runtime manage partitioning, placement, retries, and communication [4]. Modern evaluation and agent pipelines often retain the fan-out but reduce each branch to a scalar or vote. Such outputs erase four facts a reducer needs: uncertainty, information geometry, evidence identity, and inherited dependencies. Equal voting ignores the first two; repeated evidence and shared ancestry can manufacture false precision from the latter two. Snapshot and serverless systems make this failure mode operationally important because branching prepared state is increasingly convenient [1, 14, 5].

We ask a systems question: *what should a worker return so that a runtime can reduce its result without erasing the evidence behind it?* Our answer is a structured worker record and a mergeable canonical state. The current implementation solves the independent common-target case, rejects literal reuse of caller-supplied evidence identifiers, and transports lineage for later analysis. It does not infer dependence from lineage; making that inference is the research destination.

Contributions.

- **Interface.** An evidence-carrying worker record and a fixed-dimensional Gaussian numeric state for flat, streaming, and tree reduction, with provenance aggregation and exact duplicate-ID rejection.
- **Artifact.** A reference implementation with record validation, closed-form product-integral evaluation, Cholesky solves, deterministic statistical checks, and a snapshot-backed integration trace.
- **Boundary and agenda.** A precise separation between execution isolation and evidential independence, exposing fork-DAG correlation, confidence calibration, adaptive selection, and forged precision as first-class reducer requirements.

2 Evidence-Aware Worker Results

Wire record. Worker k emits

$$r_k = (\hat{\theta}_k, J_k, n_k, \mathcal{E}_k, \mathcal{L}_k, m_k), \tag{2}$$

where $\hat{\theta}_k \in \mathbb{R}^p$ is an estimate, J_k is estimated per-observation information, n_k is a literal sample count, $\mathcal{E}_k$ contains evidence identifiers, $\mathcal{L}_k$ records fork lineage, and m_k stores execution metadata. The Python dataclass has a JSON-compatible serialized form. At ingestion, the implementation rejects non-finite estimates, non-positive integer counts, malformed provenance, inconsistent dimensions, and non-symmetric or non-positive-definite J_k. Positive definiteness is a contract restriction, not an algebraic necessity: semidefinite local factors would be valid when their aggregate precision is full rank.

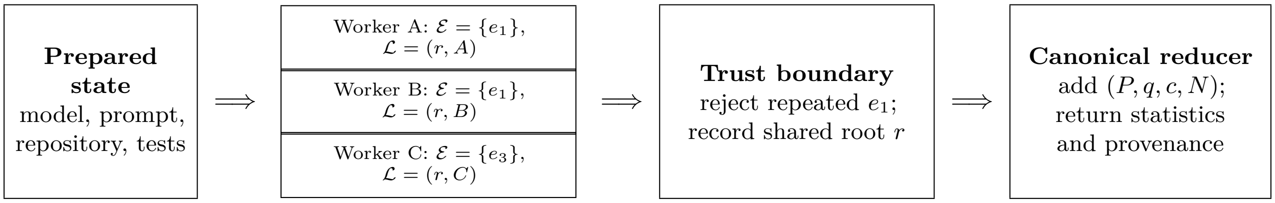

Figure 1: Execution isolation is not evidence independence. Separate workers A and B declare the same evidence e_1, which the exact-overlap guard rejects. Shared ancestry is transported even when identifiers differ, but the prototype does not yet turn that lineage into a correlation model.

The invariant quantity is total precision

$$P_k = n_k J_k, \qquad q_k = P_k \hat{\theta}_k, \qquad c_k = \hat{\theta}_k^T P_k \hat{\theta}_k. \tag{3}$$

The decomposition $P_k = n_k J_k$ is useful when n_k is an actual observation count; workers with bootstrap, clustered, weighted, or verifier-derived uncertainty may instead require a direct precision field and a separate evidence-mass convention.

Trust semantics. Evidence identifiers and lineage are optional caller-supplied labels, not verified identities. The reducer rejects overlap between nonempty $\mathcal{E}_k$ sets by default. Empty identifiers, different labels for the same data, and shared noise without literal reuse remain undetected. The current record also lacks a validated estimand, unit, model-version, or coordinate identifier, so equal vector dimension does not establish a common target. A production runtime must bind those fields to authenticated data and model manifests.

Numeric merge and provenance. Independent summaries add

$$(P, q, c, N) = \bigoplus_k (P_k, q_k, c_k, n_k). \tag{4}$$

In exact arithmetic the numeric state is associative and commutative, enabling arbitrary tree reduction. The scalar c must survive intermediate merges: re-emitting only $(\hat{\theta}, J, N)$ would erase disagreement accumulated below the tree. Provenance follows its own semantics. Evidence IDs are canonicalized as a set; lineage uses order-preserving deduplication, so the full serialized summary is intentionally not commutative. The finalized result now returns both provenance fields. Numeric state costs $O(p^2)$, whereas an exact provenance ledger can grow as $O(|\mathcal{E}| + |\mathcal{L}|)$.

Relationship to sufficient-statistic reduction. Model-specific sufficient statistics remain preferable when available: a mean is recovered from $(\sum_i x_i, n)$ and ordinary least squares from $(X^T X, X^T y)$. The contract targets one-shot settings where workers expose estimates and calibrated uncertainty but raw data or model-specific sufficient statistics are unavailable or undesirable to transmit.

Execution substrate. The record is independent of processes, containers, and microVMs. The local backend uses Python's platform-dependent process pool. The documented islo workflow prepares and saves a named environment before the normal backend restores one sandbox per shard. This separation lets an execution planner choose placement and tree shape without changing the statistical contract.

Retries and speculative execution. Evidence identity belongs to an evaluation, not to the process that happened to execute it. A retry after failure and a speculative duplicate should therefore reuse the original evidence token; accepting both would turn fault tolerance into double counting. The current overlap guard makes a repeated token detectable by raising an error; it does not implement idempotent admission. A production keyed admission layer should retain one result for retries or speculative duplicates. A branch that draws a genuinely new sample receives a new token while retaining its parent lineage. Production tokens should bind the input partition, evaluator, model and prompt versions, and selection context rather than relying on free-form strings.

3 Associative Gaussian Reduction

Statistical scope. Confidence distributions and inverse-information methods combine evidence from independent sources [12, 16, 17]; Rao-CD places estimating-function inference in a MapReduce setting [19]. We use their Gaussian/Wald common-effect special case. Assume disjoint evidence with independent estimation noise, a common parameter θ_0, fixed p and K, $\min_k n_k \to \infty$, and calibrated positive-definite precision estimates. Write V_k for worker k's per-observation asymptotic covariance and suppose

$$n_k^{1/2}(\hat{\theta}_k - \theta_0) \xrightarrow{d} N_p(0, V_k), \qquad J_k \xrightarrow{p} V_k^{-1}. \tag{5}$$

For efficient likelihood workers, local asymptotic normality is one route to this display; for estimating-equation workers, V_k is a sandwich covariance and V_k^{-1} is Godambe precision [7]. The leading Wald confidence kernel is

$$g_k(\theta) = \exp\left\{ -\tfrac{1}{2} (\theta - \hat{\theta}_k)^T P_k (\theta - \hat{\theta}_k) \right\}. \tag{6}$$

It is exact when the local estimate is Gaussian with fixed known covariance. Plug-in covariance and non-Gaussian errors return the interpretation to first-order asymptotics.

Pooled center and heterogeneity. Let $P = \sum_k P_k$, $q = \sum_k q_k$, and $c = \sum_k c_k$. Completing the square gives

$$\hat{\theta} = P^{-1}q, \qquad \Sigma_g = P^{-1}, \tag{7}$$

where Σ_g is exactly the covariance of the normalized product kernel and, under Eq. (5), a first-order estimate of the sampling covariance of $\hat{\theta}$. The residual quadratic is

$$\begin{aligned} \Delta &= \sum_k (\hat{\theta}_k - \hat{\theta})^T P_k (\hat{\theta}_k - \hat{\theta}) \\ &= c - q^T P^{-1} q \geq 0 \quad \text{in exact arithmetic.} \end{aligned} \tag{8}$$

For $p = 1$, Δ is Cochran's Q; the vector expression is the corresponding Gaussian weighted-residual diagnostic with direct multivariate meta-analytic analogues [3, 8]. Under independent $N_p(\theta_0, P_k^{-1})$ estimates with fixed known P_k, $\Delta \sim \chi^2_{p(K-1)}$; plug-in precision makes this calibration asymptotic. A small value does not prove independence or exclude duplicated evidence, and raw values are not comparable across different K without accounting for degrees of freedom.

For the unit-height kernels in Eq. (6), the partition integral is

$$Z_g = (2\pi)^{p/2} |P|^{-1/2} \exp(-\Delta/2). \tag{9}$$

Thus $\Delta/2$ is both the minimized summed quadratic energy and the heterogeneity contribution to $-\log Z_g$. The full integral also depends on $|P|$ and the integration measure, so it is neither a standalone conflict score nor model evidence. Normalized local kernels require their own Gaussian constants.

Numerics and naming. The implementation obtains $\hat{\theta}$, Σ_g, and $\log |P|$ from a Cholesky factor rather than explicit matrix inversion. It evaluates Eq. (9) in closed form, clips a negative computed Δ only within a scale-aware roundoff tolerance, and rejects larger violations of the canonical invariant; the subtraction can otherwise suffer cancellation at large offsets. Writing

$$E_k(\theta) = \tfrac{1}{2}(\theta - \hat{\theta}_k)^T J_k (\theta - \hat{\theta}_k), \qquad \beta_k = n_k \tag{10}$$

makes $g_k = \exp\{-\beta_k E_k\}$ an unnormalized Gibbs kernel. This is a naming convention: n_k can be absorbed into J_k, and P_k is the invariant. A global $\beta = N = \sum_k n_k$ with weighted-average energy is the same product. Bounded local sample sizes do not validate Gaussian factors, and accumulated local bias can dominate when K grows [18, 19].

Precision is not robustness. Holding reported precisions fixed, worker o has local influence

$$\frac{\partial \hat{\theta}}{\partial \hat{\theta}_o} = P^{-1} P_o. \tag{11}$$

Low relative precision in a direction limits leverage there, but influence does not decay with distance; fabricated high precision adds a second attack surface. The artifact's stress-test clip caps determinant-based precision volume and applies a coordinate-wise median/MAD redescending location weight. A determinant cap does not bound the largest directional eigenvalue, and the location rule is not rotation invariant. The method instruments one failure mode; it is not a Byzantine guarantee.

4 Evaluation of the Reduction Contract

The evaluation separates unit-tested identities, seeded behavioral checks, and execution traces. All statistical data are synthetic. Table 2 states the conclusion supported by each check rather than treating point gaps as an efficiency study.

Algebra and local estimators. Twelve homoscedastic mean shards of size 107–355 produce 4.9542 with a nominal plug-in 95% Wald interval $[4.8980, 5.0104]$; the full-sample mean is 4.9546. The reducer agrees with closed-form inverse-variance pooling to floating-point precision, and flat and tree reductions agree. This checks algebra; $(\sum x, n)$ would recover the sample mean exactly. In the linear-regression check, the pooled coefficient has norm gap 0.033 to centralized OLS. Each worker estimates its own residual variance, whereas exact aggregation of $X^T X$ and $X^T y$ uses a different centralized variance treatment.

Nonlinear and adversarial checks. Five IID logistic shards have sizes $\{60, 120, 400, 2000, 5000\}$. Across eight fixed seeds, information pooling is 0.0083 ± 0.0042 from the centralized MLE, while equal coefficient averaging is 0.177 ± 0.090 away. This is a sanity check against a deliberately weak baseline, not a comparison with sample-size weighting, Rao-CD, surrogate likelihood, or distributed Newton. In the forged-precision check, one scalar worker reports a distant location from a 2,000-point shard and inflates its reported precision by a further 50×; unprotected pooling moves to 17.0004, and the stress clip returns 4.9566. The trace exposes both the attack surface and the heuristic's narrow scope.

Execution paths. One integration trace starts from a named 141 MB islo snapshot and restores four sandboxes. Their seed-pinned workers return pooled mean 4.9422 versus full-sample 4.9450; four concurrent restore–run–capture round trips complete in 6.70 s. This validates the snapshot-to-worker-to-reducer path with client-observed timing.

The batched measurements in Table 1 characterize three exercised API paths with different operation boundaries; they are not a cross-platform latency benchmark.

The raw logs also retain failed exploratory sweeps: Daytona at requested client concurrency 32 and islo at 16 and 32 encountered validation or capacity walls. These failures are capacity observations, not points on a common performance curve.

Table 1: Platform-specific batched traces. Daytona and Tensorlake percentiles cover sandbox creation; their total wall time covers the complete create–run–delete batch. The islo percentiles cover restore–run–capture round trips.

Platform	State source	Measured operation	Client conc.	Batch	Success	p_{50}	p_{95}	Total wall
Daytona	default environment	create	8	1024	1024/1024	0.20 s	1.12 s	190.3 s
Tensorlake	default environment	create	8	256	256/256	3.44 s	9.00 s	387.3 s
islo	named 141 MB snapshot	restore + run + capture	12	256	255/256	6.87 s	9.04 s	240.3 s

Table 2: Claim-traceable artifact checks. The logistic values are mean ± SD over eight seeds.

Check	Observation	Supported conclusion
Canonical algebra	Flat/tree and closed-form results agree	Code matches the stated Gaussian formulas
Evidence overlap	Repeated nonempty ID raises an error	Exact labeled reuse is blocked
Unequal-size logistic	pool gap 0.0083±0.0042; equal average 0.177±0.090	Expected direction versus a deliberately weak baseline
Forged precision	attacked 17.0004; clipped 4.9566	Vulnerability and one test heuristic
Snapshot integration	four workers complete in 6.70 s	Named-snapshot path executes end to end

5 Assumptions and Open Problems

Dependence. Literal ID deduplication addresses only exact declared reuse. A lineage union records common ancestors but does not encode a fork DAG or map ancestry to covariance. A correlation-aware reducer should combine factors conditionally on shared observations, evaluators, model state, and selection history. Equation (1) shows why this is central rather than a bookkeeping refinement.

From lineage to a fork-DAG reducer. A useful next record would name parent branch IDs, immutable evidence manifests, model/prompt/tool hashes, evaluator and test versions, and branch-selection events. The runtime could then map known overlap to shared latent factors, covariance blocks, or conservative evidence groups before applying a generalized reducer. Crucially, the interface must permit abstention: when the runtime cannot justify a dependence model, it should report unresolved correlation rather than manufacture a narrower interval. The current lineage union is the transport layer for this design, not the design itself.

Different local targets. The common-θ_0 assumption defines a common-effect model. Non-IID observations do not by themselves invalidate pooling when every worker identifies the same parameter and reports valid uncertainty. If workers target different estimands, P^{-1} omits between-worker variation and can be severely overconfident. The system must first name the target and adopt an explicit heterogeneity model, such as site effects, meta-regression, or a random-effects hierarchy.

Confidence construction and adaptive selection. A model's verbal confidence is not automatically a calibrated standard error; calibration depends on the model, elicitation, task, and deployment distribution. Agent precision should be derived from external evidence such as held-out tests, repeated trials, independent verifiers, or an empirically calibrated outcome model. If the same evidence selects surviving branches and constructs their factors, winner's-curse bias can result; the ledger must record that reuse.

Untrusted workers and systems evidence. Robust aggregation requires a threat model over locations, covariance reports, collusion, and dimension. Controlled systems comparisons likewise require identical images and workloads, concurrent fan-out, tail latency, memory sharing, failures, cost, and statistical utility per unit time. The present artifact supplies interfaces and traces on which those studies can be built.

6 Related Work

The Gaussian reducer belongs to established confidence-distribution and common-effect meta-analysis [12, 16, 3]. Rao-CD and distributed GLM inference provide closer estimating-equation constructions [19, 13]. One-shot averaging and surrogate likelihood explore neighboring communication–statistics tradeoffs [18, 9]. Our contribution is the runtime boundary that determines when invoking such a reducer is justified.

Self-consistency marginalizes answers from diverse reasoning paths, while multiagent debate lets models exchange critiques before converging [15, 6]. These methods use redundant generations for selection; they do not establish independent evidence. Our complementary concern is provenance and uncertainty aggregation under stated dependence assumptions.

MapReduce established the programmable dataflow contract [4]; Firecracker, Catalyzer, REAP, and Faasm study lightweight isolation, startup, and reusable state [1, 5, 14, 11]. These systems make branching practical but do not supply an evidential independence model. Robust distributed learning methods such as Krum and FLAME start from explicit adversary models [2, 10]; the included stress clip has no comparable guarantee.

7 Conclusion

Cheap forks reduce execution cost; they do not increase the amount of independent evidence. The contract developed here makes an estimate, uncertainty, evidence identity, and lineage explicit, then reduces independent

Gaussian summaries through an associative numeric state while retaining provenance. The reference artifact validates this path and exposes residual heterogeneity as a first-class diagnostic. The next systems step is a reducer over the fork DAG: one that can explain why two branches count as separate evidence, model the dependencies they inherited, and remain calibrated after selection. Faster fan-out is useful; evidence-aware fan-in is what makes the result trustworthy.

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